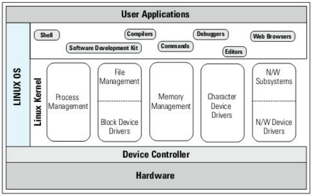

Figure 6: Schematic diagram of Linux architecture

- Carrier Grade Linux (CGL) specifications [4]
- Consumer Electronics Linux Forum (CELF) specifications [3]

Licensing issues—proprietary versus FOSS

The debate over the choice of a proprietary or open source OS is as old as computing science itself, but there are some standard parameters on which this choice should be based:

- Is time to market an important parameter for your product?
- Does your team have adequate FOSS skills?
- Are you selling your product in the volume market or niche market?
- Do you feel licence fees are more important compared to development costs?
- Do you need a custom GUI, tool-set, etc.?
- Is it fine to share your product code with others?
- Is it necessary to maintain your Intellectual Property (IP)?
- Do you want scalability for your product?
- Do you have some serious hard real-time requirements in your product, and hence want to minimise risk?

Vendor support and marketing trends

When do you need vendor support? We have to check for the presence of the following skills in the team:

- Are there adequate open source software skills in your team?
- Is there a business or technical need to use third-party software?
- Is there a need for porting from proprietary to FOSS?
- What middleware would you like to use?
- Are there real-time requirements in your solution or system?
- Are there development tools, such as a platform SDK and debugging tools for monitoring, live updating, efficient power management, security enhancement, remote debugging, etc?
- Does the OS have a minimal kernel footprint?
- What hardware support do you need?

Marketing trends

Let's understand some crucial marketing trends pertaining to the choice of platform:

- What is the market share of the platform to be selected, among the host of companies providing the platform?
- What is the roadmap or future of the OS? This will help us better understand system features, and also compatibility with long-term plans for the product.
- What kind of support does the platform have for different processor architectures, third-party products and tools?

Once we obtain details on all these aspects for the given operating system, we can decide whether the OS will satisfy our project's requirements, or whether we need to look for alternatives.

If the given operating system is Linux, then we should get a clear understanding of how Linux addresses the above aspects. The schematic diagram for Linux architecture is given in Figure 6. The split view of the Linux kernel is discussed in detail and is available at [10].

In the next article, we will explore the details about what kind of support Linux provides in the ten areas we have discussed here, beginning with file management. END

References

1. Monolithic kernel versus micro-kernel, by Benjamin Roch: *http://www.vmars.tuwien.ac.at/courses/akti12/journal/04ss/article_04ss_Roch.pdf*
2. Hybrid kernel: *http://en.wikipedia.org/wiki/Hybrid_kernel*
3. Consumer Electronic Linux Forum home page: *http://celinuxforum.org/*
4. Carrier Grade Linux home page at: *http://www.linuxfoundation.org/collaborate/workgroups/cgl*
5. Institute of Electrical and Electronics Engineers (IEEE) home page at: *http://standards.ieee.org/*
6. The Open Group: *http://www.opengroup.org/onlinepubs/009695399/*
7. Portable Operating System Interface (POSIX): *http://posixcertified.ieee.org/*
8. Linux Standard Base (LSB) specifications: *http://refspecs.freestandards.org/lsb.shtml*
9. Embedded Linux Consortium (ELC) platform specifications: *http://www.linuxfoundation.org/en/ELC*
10. Split view of Linux kernel, Figure 1-1, page 6, available at: *http://lwn.net/images/pdf/LDD3/ch01.pdf*

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